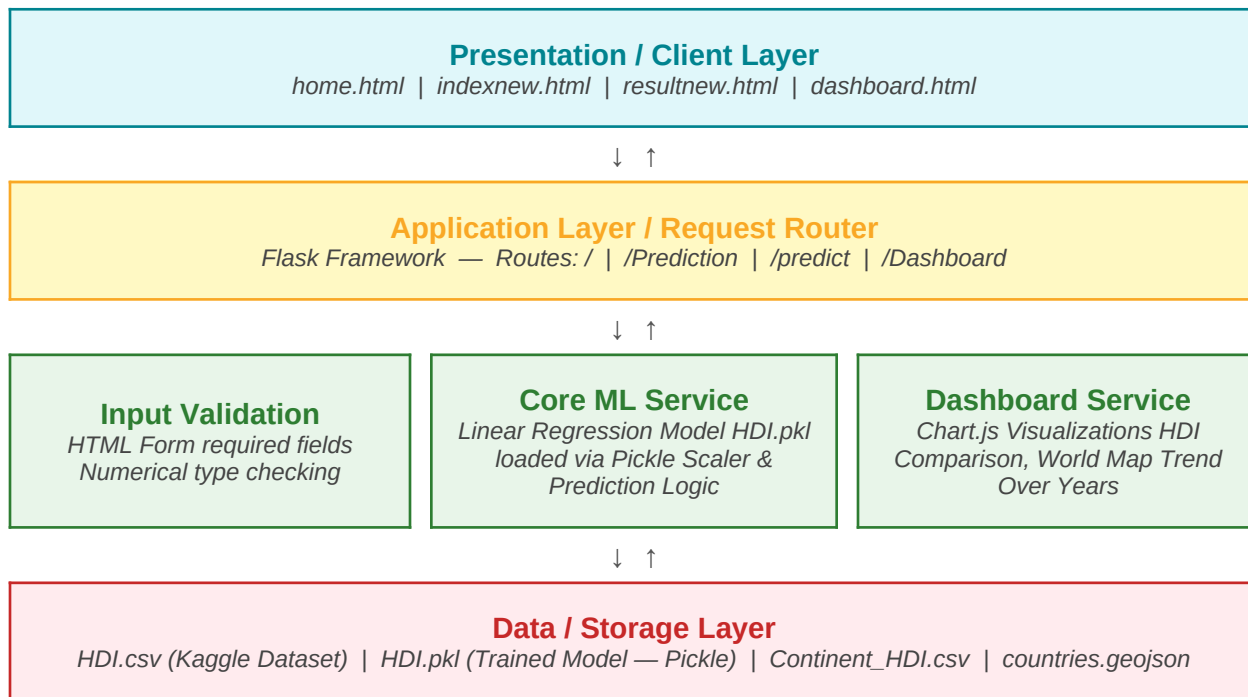


Solution Architecture

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below provides a high-level visual representation of the HDI Prediction System architecture, mapping the Presentation Layer, Application Layer, Core Services, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user-facing web interface. Includes the home page, input form, result display page, and the dashboard page with all three interactive charts	HTML5, CSS3, Chart.js, Jinja2 (Flask Templates)

Application Layer (Flask Router)	Acts as the request handler and routing engine. Receives user input from forms, processes it, calls the ML model, and returns the result to the appropriate template	Python 3.x, Flask 3.0.3
Core ML Service	The machine learning inference layer. Loads the pre-trained Linear Regression model from HDI.pkl at startup and performs HDI score prediction and category classification on incoming user inputs	Scikit-learn, Pickle, NumPy, Pandas
Dashboard Service	Renders interactive data visualizations from the HDI dataset. Provides Top 20/Bottom 10 country comparisons, category-wise distribution charts, and regional HDI trend analysis	Chart.js, HTML5, Kaggle HDI Dataset (CSV + GeoJSON)
Data / Storage Layer	Stores all project data as flat files. The HDI dataset is stored as CSV files and the trained ML model is serialized as a Pickle file, eliminating the need for a traditional database	HDI.csv, Continent_HDI.csv, countries.geojson, HDI.pkl